

### Choose Parameter Values

$r = 1$  for bias, variance

$r = 2$  for distribution, quantiles

$$c_2 = \begin{cases} 1; & \text{for } r = 1 \\ 0.1; & \text{otherwise} \end{cases}$$

### Calculate Bootstrap Estimates

(1) Choose initial block size,  $\ell = n^{\frac{1}{r+4}}$ , where  $n$  is th sample size

(2) Calculate MBB estimators,  $\hat{\varphi}_n(\ell)$  and  $\hat{\varphi}_n(2\ell)$

### Estimate $\widehat{\text{VAR}}$

(1) Set  $m = c_2 n^{1/3} \ell^{2/3}$  (no. of blocks to be deleted)

(2) Calculate the Jackknife-after-Bootstrap (JAB) Variance estimator  $\widehat{\text{VAR}}$  using  $\hat{\varphi}_n(\ell)$

(i) Collect the indices set:

$$I_i^* = \{k: 1 \leq k \leq K, \{ {}_k B_1^*, \dots, {}_k B_b^* \} \cap \{B_i, \dots, B_{i+m-1}\} = \emptyset\}$$

(ii)  $\hat{\varphi}_n^{(i)}$ :  $i$ -th block-deleted jackknife point value, calculated using  $k \in I_i^*$

$$(iii) \tilde{\varphi}_n^{(i)} = m^{-1} \left[ N \hat{\varphi}_n - (N - m) \hat{\varphi}_n^{(i)} \right]$$

(iv)  $\widehat{\text{VAR}}_{\text{JAB}}[\hat{\varphi}_n(\ell)] = \frac{m}{N-m} \frac{1}{M} \sum_{i=1}^M (\tilde{\varphi}_n^{(i)} - \hat{\varphi}_n)^2$ ,  
where,  $N$  = No. of all possible blocks,  
and  $M = N - m + 1$

### Calculate $\hat{C}_1$ and $\hat{C}_2$

$$\hat{C}_1 = n \ell^{-r} \widehat{\text{VAR}}_{\text{JAB}}[\hat{\varphi}_n(\ell)]$$

$$\hat{C}_2 = 2\ell \{ \hat{\varphi}_n(\ell) - \hat{\varphi}_n(2\ell) \}$$

### Calculate Optimal Block Length, $\hat{\ell}^0$

$$\hat{\ell}^0 = \left[ \frac{2\hat{C}_2^2}{r\hat{C}_1} \right]^{\frac{1}{r+2}} n^{\frac{1}{r+2}}$$